

第 7 章 偏微分方程数值解

本章主要内容

- (1) 抛物型方程的有限差分方法
古典显格式, 古典隐格式, Richardson 格式, Crank-Nicolson 格式
- (2) 差分格式的收敛性和稳定性
- (3) 双曲型方程的有限差分格式
显格式, 隐格式
- (4) 椭圆型方程的差分格式
五点差分格式, 收敛性

偏微分方程一般可分为抛物型方程、双曲型方程和椭圆型方程三种类型. 如下面是 3 个典型的定解问题:

(1) 一维抛物型方程初边值问题

$$\begin{cases} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f(x, t), \\ u(x, 0) = \varphi(x), \\ u(0, t) = \alpha(t), \quad u(l, t) = \beta(t), \end{cases}$$

发展方程 (时间 t)

反应扩散方程. (reaction - diffusion)
 $x \in (0, l), t \in (0, T)$

初始值.

$$x \in [0, l],$$

$$t \in [0, T].$$

(2) 一维双曲型方程初边值问题

第一型 Dirichlet

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = f(x, t), \\ u(x, 0) = \varphi(x), \quad u_t(x, 0) = \psi(x), \\ u(a, t) = \alpha(t), \quad u(b, t) = \beta(t), \end{cases}$$

$$x \in (a, b), t \in (0, T),$$

$$x \in [a, b],$$

$$t \in [0, T].$$

(3) 二维 Poisson 方程边值问题

$$\begin{cases} - \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = f(x, y), & (x, y) \in \Omega \subset \mathbf{R}^2, \\ u(x, y) = \varphi(x, y). & (x, y) \in \Gamma \end{cases}$$

Taylor 展开

用差分方法求微分方程近似解时常要用到以下几个公式:

$$\underline{g'(x_0)} = \frac{1}{h}[g(x_0 + h) - g(x_0)] - \underline{\frac{h}{2}g''(\xi_1)}, \quad \xi_1 \in (x_0, x_0 + h), \quad (0.1)$$

$$g'(x_0) = \frac{1}{h}[g(x_0) - g(x_0 - h)] + \frac{h}{2}g''(\xi_2), \quad \xi_2 \in (x_0 - h, x_0), \quad (0.2)$$

$$\underline{g'(x_0)} = \frac{1}{h} \left[g \left(\overset{t_{k+1}}{\underbrace{x_0 + \frac{h}{2}}_{h=\bar{c}}} \right) - g \left(\underbrace{x_0 - \frac{h}{2}} \right) \right] - \frac{\overset{h^2}{\underbrace{h^2}}}{24} g'''(\xi_3),$$
$$\xi_3 \in \left(x_0 - \frac{h}{2}, x_0 + \frac{h}{2} \right), \quad (0.3)$$

$$\underline{g''(x_0)} = \frac{1}{h^2}[g(x_0 + h) - 2g(x_0) + g(x_0 - h)] - \frac{h^2}{12}g^{(4)}(\xi_4),$$
$$\xi_4 \in (x_0 - h, x_0 + h), \quad (0.4)$$

$$\underline{g(x_0)} = \frac{1}{2}[g(x_0 - h) + g(x_0 + h)] - \frac{h^2}{2}g''(\xi_5),$$
$$\xi_5 \in (x_0 - h, x_0 + h). \quad \text{平均} \quad (0.5)$$

1 抛物型方程的差分方法

考虑下面的定解问题:

$$f=0 \quad \text{等}$$

$$\begin{cases} \frac{\partial u}{\partial t} - a \frac{\partial^2 u}{\partial x^2} = f(x, t), & x \in (0, l), t \in (0, T), \end{cases} \quad (1.1)$$

$$\begin{cases} u(x, 0) = \varphi(x), & x \in [0, l], \end{cases} \quad (1.2)$$

$$\begin{cases} u(0, t) = \alpha(t), & u(l, t) = \beta(t), & t \in [0, T], \end{cases} \quad (1.3)$$

其中 a 是正常数, $f, \varphi, \alpha, \beta$ 是已知函数, 满足 $\varphi(0) = \alpha(0), \varphi(l) = \beta(0)$.
假设定解问题 (1.1)-(1.3) 存在唯一解 $u(x, t)$, 且具有一定的光滑性. $\checkmark$

$$(1.3) \text{ 令 } t=0,$$

$$(1.2)$$

相容性条件.

$$\begin{cases} u(0, 0) = \alpha(0) & x=0 \quad u(0, 0) = \varphi(0) \\ u(l, 0) = \beta(0) & x=l \quad u(l, 0) = \varphi(l) \end{cases}$$

$$\Rightarrow \alpha(0) = \varphi(0), \quad \beta(0) = \varphi(l).$$

1.1 网格剖分

记 $D = \{(x, t) | 0 \leq x \leq l, 0 \leq t \leq T\}$, $h = l/M$, $\tau = T/N$, $x_i = ih$ ($0 \leq i \leq M$), $t_k = k\tau$ ($0 \leq k \leq N$), 用两簇平行直线

$$x_0 = 0$$

$$x = x_i, \quad i = 0, 1, \dots, M,$$

$$x_M = Mh = M \frac{l}{M} = l \quad t = t_k, \quad k = 0, 1, \dots, N$$

将区域 D 分割成矩形网格(见图1.1). h 和 τ 称为空间步长和时间步长. 网格点 (x_i, t_k) 称为节点. 记 $\bar{D}_h = \{(x_i, t_k) | 0 \leq i \leq M, 0 \leq k \leq N\}$, $D_h = \{(x_i, t_k) | 1 \leq i \leq M-1, 1 \leq k \leq N\}$, $\Gamma_h = \bar{D}_h \setminus D_h$. 称 D_h 为内部节点, Γ_h 为边界节点. 在位于 $t = t_k$ 上的所有节点称为第 k 层节点.

1.2 古典显格式 (向前 Euler 格式)

考虑在点 (x_i, t_k) 处的方程

$$\frac{\partial u}{\partial t}(x_i, t_k) - a \frac{\partial^2 u}{\partial x^2}(x_i, t_k) = f(x_i, t_k). \quad (1.4)$$

由 (0.1) 和 (0.4) 可得

$$\frac{\partial u}{\partial t}(x_i, t_k) = \frac{1}{\tau} [u(x_i, t_{k+1}) - u(x_i, t_k)] - \frac{\tau}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \eta_i^k),$$

t_{k+1}

$\eta_i^k \in (t_k, t_{k+1}),$

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_k) = \frac{1}{h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)] - \frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k), \quad \xi_i^k \in (x_{i-1}, x_{i+1}).$$

将上面两式代入 (1.4) 得

$$\frac{1}{\tau} [u(x_i, t_{k+1}) - u(x_i, t_k)] - \frac{a}{h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)]$$

$$\underline{U_i^k = u(x_i, t_k)}$$

$$u(x_i, t_0) = \varphi(x_i) \quad 0 \leq i \leq M, \quad u(x_0, t_k) = \alpha(t_k) \quad u(x_M, t_k) = \beta(t_k) \quad 1 \leq k \leq N$$

$$= f(x_i, t_k) + \frac{\tau}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \eta_i^k) - \frac{ah^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k). \quad (1.5)$$

$$1 \leq i \leq M-1, \quad 0 \leq k \leq N-1.$$

由初始条件 (1.2) 和边界条件 (1.3) 得 $i=0 \quad u(x_0, t_0) = \varphi(x_0) = \varphi(0)$

$$t_0=0 \quad u(x_i, t_0) = \varphi(x_i) \quad 1 \leq i \leq M-1, \quad u(x_M, t_0) = \varphi(x_M) = \varphi(l)$$

$$u(x_0, t_k) = \alpha(t_k), \quad u(x_M, t_k) = \beta(t_k), \quad 0 \leq k \leq N.$$

记

$$O(\tau+h^2) = R_{ik}^{(1)} = \frac{\tau}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \eta_k) - \frac{ah^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i, t_k).$$

$k=0$ 时 $u(x_0, t_0) = \alpha(t_0) = \alpha(0) = \varphi(0)$
 $u(x_M, t_0) = \beta(t_0) = \beta(0) = \varphi(l)$
 相符合

在 (1.5) 中忽略 $R_{ik}^{(1)}$, 并用 u_i^k 代替 $u(x_i, t_k)$ 得下面的差分格式:

$$\begin{cases} \frac{1}{\tau}(u_i^{k+1} - u_i^k) - \frac{a}{h^2}(u_{i+1}^k - 2u_i^k + u_{i-1}^k) = f(x_i, t_k), \\ 1 \leq i \leq M-1, \quad 0 \leq k \leq N-1, \end{cases} \quad (1.6)$$

$$u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \quad (1.7)$$

$$u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \quad (1.8)$$

差分格式 (1.6)-(1.8) 称为古典显格式 (向前 Euler 格式), $R_{ik}^{(1)}$ 称为差分格式 (1.6)-(1.8) 的截断误差. 记 $r = \frac{a\tau}{h^2}$, 称 r 是步长比. (1.6) 可写

为

$$r = \frac{a\tau}{h^2}$$

$$u_i^{k+1} = (1 - 2r)u_i^k + r(u_{i-1}^k + u_{i+1}^k) + \tau f(x_i, t_k).$$

可将差分格式 (1.6)-(1.8) 写成向量和矩阵形式:

$$\begin{bmatrix} u_1^{k+1} \\ u_2^{k+1} \\ \vdots \\ u_{M-2}^{k+1} \\ u_{M-1}^{k+1} \end{bmatrix} = \begin{bmatrix} 1-2r & r & & & \\ & r & 1-2r & r & \\ & & \ddots & \ddots & \ddots \\ & & & r & 1-2r & r \\ & & & & r & 1-2r \end{bmatrix} \begin{bmatrix} u_1^k \\ u_2^k \\ \vdots \\ u_{M-2}^k \\ u_{M-1}^k \end{bmatrix} + \begin{bmatrix} \tau f(x_1, t_k) + r\alpha(t_k) \\ \tau f(x_2, t_k) \\ \vdots \\ \tau f(x_{M-2}, t_k) \\ \tau f(x_{M-1}, t_k) + r\beta(t_k) \end{bmatrix}, \quad k = 0, 1, \dots, N-1.$$

$$i=1 \text{ 时 } u_1^{k+1} = (1-2r)u_1^k + r u_2^k + r\alpha(t_k) + \tau f(x_1, t_k)$$

$$i=M-1 \text{ 时 } u_{M-1}^{k+1} = (1-2r)u_{M-1}^k + r u_{M-2}^k + r u_M^k + \tau f(x_{M-1}, t_k)$$

1.3 古典隐格式 (向后 Euler 格式)

还是考虑 (x_i, t_k) 点的方程

$$\frac{\partial u}{\partial t}(x_i, t_k) - a \frac{\partial^2 u}{\partial x^2}(x_i, t_k) = f(x_i, t_k). \quad (1.9)$$

$\frac{\partial u}{\partial t}(x_i, t_k)$ 用向后差商 (0.2) 近似, $\frac{\partial^2 u}{\partial x^2}(x_i, t_k)$ 还是用二阶差商近似得

$$\frac{\partial u}{\partial t}(x_i, t_k) = \frac{1}{\tau} [u(x_i, t_k) - u(x_i, t_{k-1})] + \frac{\tau}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \eta_i^k),$$

$\eta_i^k \in (t_{k-1}, t_k),$

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_k) = \frac{1}{h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)]$$

$$- \frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k), \quad \xi_i^k \in (x_{i-1}, x_{i+1}).$$

代入 (1.9) 得

$$\frac{1}{\tau} [u(x_i, \underline{t_k}) - u(x_i, \underline{t_{k-1}})]$$

$$\begin{aligned}
 & -\frac{a}{h^2}[u(\underline{x_{i+1}}, t_k) - 2u(x_i, t_k) + u(\underline{x_{i-1}}, t_k)] \\
 = & f(x_i, t_k) \left[-\frac{\tau}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \eta_i^k) - \frac{ah^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k) \right] \\
 & 1 \leq i \leq M-1, \quad 1 \leq k \leq N.
 \end{aligned} \tag{1.10}$$

由初始条件 (1.2) 和边界条件 (1.3) 得

$$\begin{cases} u(x_i, 0) = \varphi(x_i) & 1 \leq i \leq M-1, \\ u(x_0, t_k) = \alpha(t_k), \quad u(x_M, t_k) = \beta(t_k), & 0 \leq k \leq N. \end{cases}$$

记

$$R_{ik}^{(2)} = -\frac{\tau}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \eta_k) - \frac{ah^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i, t_k).$$

在 (1.10) 中忽略 $R_{ik}^{(2)}$, 并用 $\underline{u_i^k}$ 代替 $\underline{u(x_i, t_k)}$ 得下面的差分格式:

$$\begin{cases} \frac{1}{\tau}(\underline{u_i^k} - \underline{u_i^{k-1}}) - \frac{a}{h^2}(\underline{u_{i+1}^k} - 2\underline{u_i^k} + \underline{u_{i-1}^k}) = f(x_i, t_k), \\ 1 \leq i \leq M-1, \quad 1 \leq k \leq N, \end{cases} \tag{1.11}$$

$$u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \tag{1.12}$$

$$\underline{u_0^k} = \underline{\alpha(t_k)}, \quad \underline{u_M^k} = \underline{\beta(t_k)}, \quad 0 \leq k \leq N. \tag{1.13}$$

$$r = \frac{\tau a}{h^2}$$

差分格式 (1.11)-(1.13) 称为古典隐格式 (向后 **Euler** 格式). 将其写成矩阵向量形式:

$$\begin{bmatrix} \underline{1 + 2r} & \underline{-r} & & & \\ -r & \underline{1 + 2r} & -r & & \\ & \ddots & \ddots & \ddots & \\ & & -r & 1 + 2r & -r \\ & & & \underline{-r} & \underline{1 + 2r} \end{bmatrix} \begin{bmatrix} u_1^k \\ u_2^k \\ \vdots \\ u_{M-2}^k \\ \underline{u_{M-1}^k} \end{bmatrix} = \begin{bmatrix} u_1^{k-1} \\ u_2^{k-1} \\ \vdots \\ u_{M-2}^{k-1} \\ \underline{u_{M-1}^{k-1}} \end{bmatrix} + \begin{bmatrix} \frac{\tau f(x_1, t_k) + r\alpha(t_k)}{\underline{\tau f(x_2, t_k)}} \\ \vdots \\ \tau f(x_{M-2}, t_k) \\ \underline{\tau f(x_{M-1}, t_k) + r\beta(t_k)} \end{bmatrix}, \quad k = 1, 2, \dots, N.$$

注意边界条件 u_0^k, u_M^k

$$u(x_i, 0) = \varphi(x_i)$$

1.4 Richardson 格式

考虑节点 (x_i, t_k) 处方程

$$\frac{\partial u}{\partial t}(x_i, t_k) - a \frac{\partial^2 u}{\partial x^2}(x_i, t_k) = f(x_i, t_k),$$

$$\frac{\partial u}{\partial t}(x_i, 0) = a \frac{\partial^2 u}{\partial x^2}(x_i, 0) = f(x_i, 0)$$

将 $\frac{\partial u}{\partial t}(x_i, t_k)$ 用中心差商近似: $\Rightarrow \frac{\partial u}{\partial t}(x_i, 0) = \frac{f(x_i, 0) + h y''(x_i)}{2}$

$$\frac{\partial u}{\partial t}(x_i, t_k) = \frac{1}{2\tau} [u(x_i, t_{k+1}) - u(x_i, t_{k-1})] - \frac{\tau^2}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i^k),$$

将 $\frac{\partial^2 u}{\partial x^2}(x_i, t_k)$ 还是用 2 阶差商近似:

$$\eta_i^k \in (t_{k-1}, t_{k+1})$$

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_k) = \frac{1}{h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)] - \frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k), \quad \xi_i^k \in (x_{i-1}, x_{i+1}).$$

然后代入方程得

$$\frac{1}{2\tau} [u(x_i, t_{k+1}) - u(x_i, t_{k-1})]$$

$$\begin{aligned}
 & -\frac{a}{h^2}[u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)] \\
 = & f(x_i, t_k) + \frac{\tau^2}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i^k) - \frac{ah^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k), \\
 & 1 \leq i \leq M-1, \quad 0 \leq k \leq N-1.
 \end{aligned}$$

记

截断误差 $R_{ik}^{(4)} = \frac{\tau^2}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i^k) - \frac{ah^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k), = O(\tau^2 + h^2)$

忽略 $R_{ik}^{(4)}$, 并注意到初边值条件得下面的 Richardson 格式 $u_i^k(x_i, t_k)$

$$\begin{cases} \frac{1}{2\tau}(u_i^{k+1} - u_i^{k-1}) - \frac{a}{h^2}(u_{i+1}^k - 2u_i^k + u_{i-1}^k) = f(x_i, t_k), \\ \quad 1 \leq i \leq M-1, \quad 1 \leq k \leq N-1, \\ u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \\ u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \end{cases}$$

Richardson 格式是一个 3 层格式, 需知道 $(k-1)$ 层和 k 层的值才能求出第 $(k+1)$ 层的值.

由方程和初边值条件得

$$u_i^1 \approx \boxed{u(x_i, \tau)} = u(x_i, 0) + \tau \boxed{\frac{\partial u}{\partial t}(x_i, 0)} + \frac{\tau^2}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \bar{\eta}_i)$$

$$= \varphi(x_i) + \tau [a\varphi''(x_i) + f(x_i, 0)] + \frac{\tau^2}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \bar{\eta}_i).$$

$\eta_i \in (0, \tau)$
 $O(\tau^2)$

因此可取第 1 层的值

$$u_i^1 = \varphi(x_i) + \tau [a\varphi''(x_i) + f(x_i, 0)], \quad 1 \leq i \leq M-1.$$

完整的 Richardson 格式

$$\left\{ \begin{array}{l} \frac{1}{2\tau}(u_i^{k+1} - u_i^{k-1}) - \frac{a}{h^2}(u_{i+1}^k - 2u_i^k + u_{i-1}^k) = f(x_i, t_k), \\ \quad 1 \leq i \leq M-1, \quad 1 \leq k \leq N-1, \quad (1.14) \\ u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \quad (1.15) \\ u_i^1 = \varphi(x_i) + \tau [a\varphi''(x_i) + f(x_i, 0)], \quad 1 \leq i \leq M-1, \quad (1.16) \\ u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \quad (1.17) \end{array} \right.$$

后面可以知道, Richardson 格式是 完全不稳定的格式, 无实用价值.

τ, h 不能取

1.5 Crank-Nicolson 格式

为了要提高时间方向的精度, 记 $t_{k+\frac{1}{2}} = t_k + \frac{\tau}{2}$, 考虑点 $(x_i, t_{k+\frac{1}{2}})$ 处的微分方程

$$\frac{\partial u}{\partial t}(x_i, t_{k+\frac{1}{2}}) - a \frac{\partial^2 u}{\partial x^2}(x_i, t_{k+\frac{1}{2}}) = f(x_i, t_{k+\frac{1}{2}}). \quad (1.18)$$

利用中心差商 (0.3)

$$\frac{\partial u}{\partial t}(x_i, t_{k+\frac{1}{2}}) = \frac{1}{\tau} [u(x_i, t_{k+1}) - u(x_i, t_k)] - \frac{\tau^2}{24} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i^k),$$

$\eta_i^k \in (t_k, t_{k+1}),$ 不是 $t_{k+\frac{1}{2}}$

由平均公式 (0.5) 和二阶差商公式得

$$\begin{aligned} \frac{\partial u}{\partial t}(x_i, t_{k+\frac{1}{2}}) - \frac{a}{2} \left[\frac{\partial^2 u}{\partial x^2}(x_i, t_k) + \frac{\partial^2 u}{\partial x^2}(x_i, t_{k+1}) \right] \\ = f(x_i, t_{k+\frac{1}{2}}) - \frac{a\tau^2}{8} \frac{\partial^4 u}{\partial x^2 \partial t^2}(x_i, \bar{\eta}_i^k), \quad \bar{\eta}_i^k \in (t_k, t_{k+1}), \end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_k) = \frac{1}{h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)]$$

$$\begin{aligned}
& -\frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k), \quad \xi_i^k \in (x_{i-1}, x_{i+1}), \\
\frac{\partial^2 u}{\partial x^2}(x_i, t_{k+1}) &= \frac{1}{h^2} [u(x_{i+1}, t_{k+1}) - 2u(x_i, t_{k+1}) + u(x_{i-1}, t_{k+1})] \\
& -\frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\tilde{\xi}_i^k, t_{k+1}), \quad \tilde{\xi}_i^k \in (x_{i-1}, x_{i+1}),
\end{aligned}$$

将上面几个式子代入 (1.18) 得

$$\begin{aligned}
& \frac{1}{\tau} [u(x_i, t_{k+1}) - u(x_i, t_k)] \\
& - \frac{a}{2h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k) \\
& + u(x_{i+1}, t_{k+1}) - 2u(x_i, t_{k+1}) + u(x_{i-1}, t_{k+1})] \\
& = \underbrace{f(x_i, t_{k+\frac{1}{2}})}_{\text{wavy}} + \frac{\tau^2}{24} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i^k) - \frac{ah^2}{24} \left[\frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k) + \frac{\partial^4 u}{\partial x^4}(\tilde{\xi}_i^k, t_{k+1}) \right] \\
& - \frac{a\tau^2}{8} \frac{\partial^4 u}{\partial x^2 \partial t^2}(x_i, \bar{\eta}_i^k), \quad 1 \leq i \leq M-1, \quad 0 \leq k \leq N-1.
\end{aligned}$$

记

$$R_{ik}^{(3)} = \frac{\tau^2}{24} \left[\frac{\partial^3 u}{\partial t^3}(x_i, \eta_i^k) - 3a \frac{\partial^4 u}{\partial x^2 \partial t^2}(x_i, \bar{\eta}_i^k) \right] \\ - \frac{ah^2}{24} \left[\frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k) + \frac{\partial^4 u}{\partial x^4}(\tilde{\xi}_i^k, t_{k+1}) \right] = O(\tau^2 + h^4)$$

略去 $R_{ik}^{(3)}$, 用 u_i^k 代替 $u(x_i, t_k)$ 并注意到初边值条件得下面的差分格式

$$\begin{cases} \frac{u_i^{k+1} - u_i^k}{\tau} - a \cdot \frac{1}{2} \left[\frac{u_{i+1}^k - 2u_i^k + u_{i-1}^k}{h^2} + \frac{u_{i+1}^{k+1} - 2u_i^{k+1} + u_{i-1}^{k+1}}{h^2} \right] \\ = f(x_i, t_{k+\frac{1}{2}}), \quad 1 \leq i \leq M-1, \quad 0 \leq k \leq N-1, \end{cases} \quad (1.19)$$

$$u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \quad (1.20)$$

$$u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \quad (1.21)$$

$= \bar{u}$

隐式

C-N 格式.

上面的差分格式称为 Crank-Nicolson 格式. 用矩阵和向量表示为

$$= \begin{bmatrix} 1+r & -\frac{r}{2} & & & \\ -\frac{r}{2} & 1+r & -\frac{r}{2} & & \\ & \ddots & \ddots & \ddots & \\ & & -\frac{r}{2} & 1+r & -\frac{r}{2} \\ & & & -\frac{r}{2} & 1+r \end{bmatrix} \begin{bmatrix} u_1^{k+1} \\ u_2^{k+1} \\ \vdots \\ u_{M-2}^{k+1} \\ u_{M-1}^{k+1} \end{bmatrix} \\ + \begin{bmatrix} 1-r & \frac{r}{2} & & & \\ \frac{r}{2} & 1-r & \frac{r}{2} & & \\ & \ddots & \ddots & \ddots & \\ & & \frac{r}{2} & 1-r & \frac{r}{2} \\ & & & \frac{r}{2} & 1-r \end{bmatrix} \begin{bmatrix} u_1^k \\ u_2^k \\ \vdots \\ u_{M-2}^k \\ u_{M-1}^k \end{bmatrix} \\ + \begin{bmatrix} \tau f(x_1, t_{k+\frac{1}{2}}) + \frac{r}{2}(\alpha(t_k) + \alpha(t_{k+1})) \\ \tau f(x_2, t_{k+\frac{1}{2}}) \\ \vdots \\ \tau f(x_{M-2}, t_{k+\frac{1}{2}}) \\ \tau f(x_{M-1}, t_{k+\frac{1}{2}}) + \frac{r}{2}(\beta(t_k) + \beta(t_{k+1})) \end{bmatrix}$$

注意这里

的系数

u_0^k, u_m^k

1.6 算例

向前 Euler

例 1.1 用古典显格式、古典隐格式、Richardson 格式和 Crank-Nicolson 格式计算下面问题的数值解：

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad x \in (0, 1), \quad t \in (0, 1], \\ u(x, 0) &= e^x, \quad x \in (0, 1), \\ u(0, t) &= e^t, \quad u(1, t) = e^{t+1}, \quad t \in [0, 1].\end{aligned}$$

该问题的精确解为 e^{x+t} .

表 1.1 给出了当 $h = 1/10$, $\tau = 1/200$ 时用古典显格式计算得到的部分数值结果, 数值解很好地逼近精确解。

表 1.1 古典显格式 $h = 1/10, \tau = 1/200$

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.00)	1.648721	1.648721	0.000000
10	(0.5, 0.05)	1.733119	1.733253	0.000134
20	(0.5, 0.10)	1.821888	1.822119	0.000230
40	(0.5, 0.20)	2.013408	2.013753	0.000344
80	(0.5, 0.40)	2.459134	2.459603	0.000469
160	(0.5, 0.80)	3.668588	3.669297	0.000708
200	(0.5, 1.00)	4.480824	4.481689	0.000865

✓ 初始值

$$a=1$$

$$r = \frac{a\tau}{h^2}$$

$$= \frac{1}{200} / \left(\frac{1}{10}\right)^2$$

$$= \frac{1}{2}$$

$$t_{10} = k\tau = 10 \times \frac{1}{200} = \underline{0.05}$$

表 1.2 给出了当 $h = 1/10$, $\tau = 1/100$ 时用古典显格式计算得到的部分数值结果, 随着计算层数的增加, 误差越来越大, 数值结果没有实用意义.

表 1.2 古典显格式 $h = 1/10, \tau = 1/100$

有条件.

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.00)	1.648721	1.648721	0.000000
4	(0.5, 0.04)	1.715721	1.716007	0.000286
8	(0.5, 0.08)	1.787434	1.786038	0.001396
12	(0.5, 0.12)	2.054643	1.858928	0.195715
16	(0.5, 0.16)	17.26231	1.934729	15.32752
17	(0.5, 0.17)	43.00026	1.954237	44.95449
18	(0.5, 0.18)	133.4547	1.973878	131.4808

$$r = \frac{1}{100} / \left(\frac{1}{10}\right)^2 = 1 > \frac{1}{2}$$

表 1.3 给出了当 $h = 1/10$, $\tau = 1/100$ 时用古典隐格式计算得到的部分数值结果, 数值解很好地逼近精确解.

表 1.3 古典隐格式 $h = 1/10, \tau = 1/100$

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.00)	1.648721	1.648721	0.000000
10	(0.5, 0.10)	1.822891	1.822119	0.000772
20	(0.5, 0.20)	2.014927	2.013753	0.001174
30	(0.5, 0.30)	2.226965	2.225541	0.001424
40	(0.5, 0.40)	2.461227	2.459603	0.001624
50	(0.5, 0.50)	2.720096	2.718282	0.001814
60	(0.5, 0.60)	3.006178	3.004166	0.002012
70	(0.5, 0.70)	3.322344	3.320117	0.002227
80	(0.5, 0.80)	3.860015	3.857426	0.002589
90	(0.5, 0.90)	4.057922	4.055200	0.002722
100	(0.5, 1.0)	4.484697	4.481689	0.003008

$O(1+h)$

10

$O(746)$

表 1.4 给出了当 $h = 1/10$, $\tau = 1/100$ 时用 Richardson 格式计算得到的部分数值结果, 随着计算层数的增加, 误差越来越大, 数值结果没有实用意义.

表 1.4 Richardson 格式 $h = 1/10$, $\tau = 1/100$

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.0)	1.648721	1.648721	0.000000
1	(0.5, 0.01)	0.116501	1.665291	1.548790
2	(0.5, 0.02)	8.837941	1.682028	7.155913
3	(0.5, 0.03)	-22.13696	1.698932	23.83589
4	(0.5, 0.04)	90.79588	1.716007	89.07987
5	(0.5, 0.05)	-329.1597	1.733253	330.8929
6	(0.5, 0.06)	1267.675	1.750672	1265.924
7	(0.5, 0.07)	-4908.295	1.768267	4910.063
8	(0.5, 0.08)	19285.74	1.786083	19283.96

表 1.5 给出了当 $h = 1/10$, $\tau = 1/10$ 时用 Crank-Nicolson 格式计算得到的部分数值结果, 数值解很好地逼近精确解.

表 1.5 Crank-Nicolson 格式 $h = 1/10$, $\tau = 1/10$

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.0)	1.648721	1.648721	0.000000
1	(0.5, 0.1)	1.822394	1.822119	0.000231
2	(0.5, 0.2)	2.014105	2.013753	0.000352
3	(0.5, 0.3)	2.225953	2.225541	0.000412
4	(0.5, 0.4)	2.460072	2.459603	0.000469
5	(0.5, 0.5)	2.718802	2.718282	0.000520
6	(0.5, 0.6)	3.004743	3.004166	0.000577
7	(0.5, 0.7)	3.320755	3.320117	0.000638
8	(0.5, 0.8)	3.670002	3.669297	0.000705
9	(0.5, 0.9)	4.055979	4.055200	0.000779
10	(0.5, 1.0)	4.482550	4.481689	0.000861

0.17467

2 差分格式的稳定性和收敛性

对抛物方程定解问题

热传导方程.

$$\begin{cases} \frac{\partial u}{\partial t} - a \frac{\partial^2 u}{\partial x^2} = f(x, t), & x \in (0, l), t \in (0, T), \\ u(x, 0) = \varphi(x), & x \in [0, l], \\ u(0, t) = \alpha(t), & u(l, t) = \beta(t), \quad t \in [0, T] \end{cases}$$

建立了四个差分格式.

$$y(0) = \alpha(0), \quad y(1) = \beta(0)$$

古典显格式 (向前 Euler 格式)

$$\begin{cases} \frac{1}{\tau}(u_i^{k+1} - u_i^k) - \frac{a}{h^2}(u_{i+1}^k - 2u_i^k + u_{i-1}^k) = f(x_i, t_k), \\ \quad 1 \leq i \leq M-1, \quad 0 \leq k \leq N-1, \\ u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \\ u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \end{cases}$$

$$\begin{cases} \frac{1}{\tau}(v_i^{k+1} - v_i^k) - \frac{a}{h^2}(v_{i+1}^k - 2v_i^k + v_{i-1}^k) = f(x_i, t_k) \\ v_i^0 = \varphi(x_i) \\ v_0^k = \alpha(t_k), \quad v_M^k = \beta(t_k) \end{cases}$$

$$\Sigma_i^k = u_i^k - v_i^k$$

古典隐格式 (向后 Euler 格式)

$$\begin{cases} \frac{1}{\tau}(\Sigma_i^{k+1} - \Sigma_i^k) - \frac{a}{h^2}(\Sigma_{i+1}^k - 2\Sigma_i^k + \Sigma_{i-1}^k) = 0 \\ \Sigma_i^0 = \varphi(x_i) - \varphi(x_i), \quad \Sigma_0^k = \Sigma_M^k = 0 \end{cases}$$

$$\begin{cases} \frac{1}{\tau}(u_i^k - u_i^{k-1}) - \frac{a}{h^2}(u_{i+1}^k - 2u_i^k + u_{i-1}^k) = f(x_i, t_k), \\ 1 \leq i \leq M-1, \quad 1 \leq k \leq N, \\ u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \\ u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \end{cases}$$

Richardson 格式

理查森格式

$$\begin{cases} \frac{1}{2\tau}(u_i^{k+1} - u_i^{k-1}) - \frac{a}{h^2}(u_{i+1}^k - 2u_i^k + u_{i-1}^k) = f(x_i, t_k), \\ 1 \leq i \leq M-1, \quad 1 \leq k \leq N-1, \\ u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \\ u_i^1 = \varphi(x_i) + \tau[a\varphi''(x_i) + f(x_i, 0)], \quad 1 \leq i \leq M-1, \\ u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \end{cases}$$

EN格式:

$$u_i^k \rightarrow u(x_i, t_k) \quad (\tau, h \rightarrow 0)$$

$$\begin{cases} \frac{u_i^{k+1} - u_i^k}{\tau} - a \cdot \frac{1}{2} \left[\frac{u_{i+1}^k - 2u_i^k + u_{i-1}^k}{h^2} + \frac{u_{i+1}^{k+1} - 2u_i^{k+1} + u_{i-1}^{k+1}}{h^2} \right] \\ = f(x_i, t_{k+\frac{1}{2}}), \quad 1 \leq i \leq M-1, \quad 0 \leq k \leq N-1, \\ u_i^0 = \varphi(x_i), \quad 1 \leq i \leq M-1, \\ u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \end{cases}$$

称古典显(隐)格式和 Crank-Nicolson 格式为两层格式, 称 Richardson 格式为三层格式.

对于差分格式, 我们需要讨论稳定性和收敛性.

稳定性是考虑的计算过程中的误差传播问题; 收敛性考虑的是当步长趋于零时差分方程的解是否趋于微分方程问题的解.

我们需要引进范数作为工具.

记 $\Omega_h = (x_0, x_1, \dots, x_M)$. 称 $w = (w_0, w_1, \dots, w_M)$ 为 Ω_h 上的网格函数. 记 $\mathbb{W} = \{w \mid w \text{ 为 } \Omega_h \text{ 上的网格函数}\}$. 称 $\mathbb{W}$ 为网格函数空间.

$$C[a, b] = \{f \mid f \text{ 为 } [a, b] \text{ 上的连续函数}\}$$

$n \sim 10^6$
 $x \in [a, b]$ f

对于网格函数引进范数。设 $w \in \mathbb{W}$. 定义下面的范数:

L_2 -范数 $\|w\|_2 = \sqrt{h \left[\frac{1}{2} (w_0)^2 + \sum_{i=1}^{M-1} (w_i)^2 + \frac{1}{2} (w_M)^2 \right]}.$

L_∞ -范数 $\|w\|_\infty = \max_{0 \leq i \leq M} |w_i|.$

L_1 -范数 $\|w\|_1 = h \left[\frac{1}{2} |w_0| + \sum_{i=1}^{M-1} |w_i| + \frac{1}{2} |w_M| \right].$

$\int_0^L w^2 dx$

$\int_0^L |w| dx$

设 $\{u_i^k \mid 0 \leq i \leq M, 0 \leq k \leq N\}$ 是差分格式的解. 定义

$\downarrow$
 x

$u^k = (u_0^k, u_1^k, \dots, u_M^k),$

$\Omega_h = (x_0, \dots, x_M).$

则 u^k 为 Ω_h 上的网格函数。

$0 \leq k \leq N.$

$u^0 = (u_0^0, u_1^0, \dots, u_M^0)$

$u^N = (u_0^N, u_1^N, \dots, u_M^N)$

定义 2.1 设 $\{u_i^k | 0 \leq i \leq M, 0 \leq k \leq N\}$ 是差分格式的解, $\{v_i^k | 0 \leq i \leq M, 0 \leq k \leq N\}$ 是由于 初始数据有误差 而得到的差分格式的近似解, 记

$$\varepsilon_i^k = u_i^k - v_i^k, \quad 0 \leq i \leq M, 0 \leq k \leq N.$$

如果存在与步长 h, τ 无关的常数 C , 使得

$\xi^k = (\xi_0^k, \xi_1^k, \dots, \xi_M^k) \quad \|\varepsilon^k\| \leq C \|\varepsilon^0\|, \quad 1 \leq k \leq N, \quad \text{(两层格式)}$ 古典显格式
或者 C-N

$\|\varepsilon^k\| \leq C(\|\varepsilon^0\| + \|\varepsilon^1\|), \quad 1 \leq k \leq N, \quad \text{(三层格式)}$

则称该差分格式关于范数 $\|\cdot\|$ 是稳定的, 否则称不稳定.

我们有下面的结论:

$r = \frac{\Delta t}{h^2}$

条件稳定的.

定理 2.1 当步长比 $r \leq \frac{1}{2}$ 时, 古典显格式关于 L_∞ 范数是稳定的;
当 $r > \frac{1}{2}$ 关于 L_∞ 范数不稳定.

无条件稳定.

定理 2.2 对任意步长比 r , 古典隐格式关于 L_∞ 范数是稳定的.

定理 2.3 对任意步长比 r , Crank-Nicolson 格式关于 L_2 范数稳定.

定理 2.4 对任意步长比 r , Richardson 格式关于 L_∞ 范数和 L_2 范数都是不稳定的.

完全不稳定.

定义 2.2 设 $U_i^k = u(x_i, t_k)$ 是微分方程定解问题的解, u_i^k 是对应的差分格式的解. 记

$$e_i^k = U_i^k - u_i^k, \quad 0 \leq i \leq M, 0 \leq k \leq N.$$

如果

$$\lim_{\substack{h \rightarrow 0 \\ \tau \rightarrow 0}} \max_{0 \leq k \leq N} \|e^k\| = 0,$$

则称差分格式在范数 $\|\cdot\|$ 下是收敛的. 如果

$$\max_{0 \leq k \leq N} \|e^k\| = O(h^p + \tau^q),$$

则称差分格式关于空间步长 p 阶、关于时间步长 q 阶收敛.

定理 2.5 当步长比 $r \leq \frac{1}{2}$ 时, 古典显格式在 L_∞ 范数下关于空间步长 2 阶、关于时间步长 1 阶收敛的.

定理 2.6 对任意步长比 r , 古典隐格式在 L_∞ 范数下关于空间步长 2 阶、关于时间步长 1 阶收敛的.

定理 2.7 对任意步长比 r , Crank-Nicolson 格式在 L_2 范数下关于空间步长和时间步长都是 2 阶收敛的.

$$e^k = (e_0^k, e_1^k, \dots, e_m^k)$$

无抖动

达到同阶精度. C-N 时间步长

3 双曲型方程的差分方法

考虑下面的初边值问题：

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t), & 0 < x < l, \quad 0 < t < T, \\ \underline{u(x, 0) = \varphi(x), \quad u_t(x, 0) = \psi(x)}, & 0 < x < l \\ \underline{u(0, t) = \alpha(t), \quad u(l, t) = \beta(t)}, & 0 \leq t \leq T. \end{cases}$$

和上节一样, 记 $h = l/M$, $\tau = T/N$, $x_i = ih$, $t_k = k\tau$.

3.1 显格式

考虑 (x_i, t_k) 处的方程

$$\frac{\partial^2 u}{\partial t^2}(x_i, t_k) - a^2 \frac{\partial^2 u}{\partial x^2}(x_i, t_k) = f(x_i, t_k) \quad (3.1)$$

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2}(x_i, t_k) = & \frac{1}{\tau^2} [u(x_i, \underline{t_{k+1}}) - 2u(x_i, \underline{t_k}) + u(x_i, \underline{t_{k-1}})] \\ & - \frac{\tau^2}{12} \frac{\partial^4 u}{\partial t^4}(x_i, \eta_i^k), \quad \eta_i^k \in (\underline{t_{k-1}}, \underline{t_{k+1}}), \end{aligned}$$

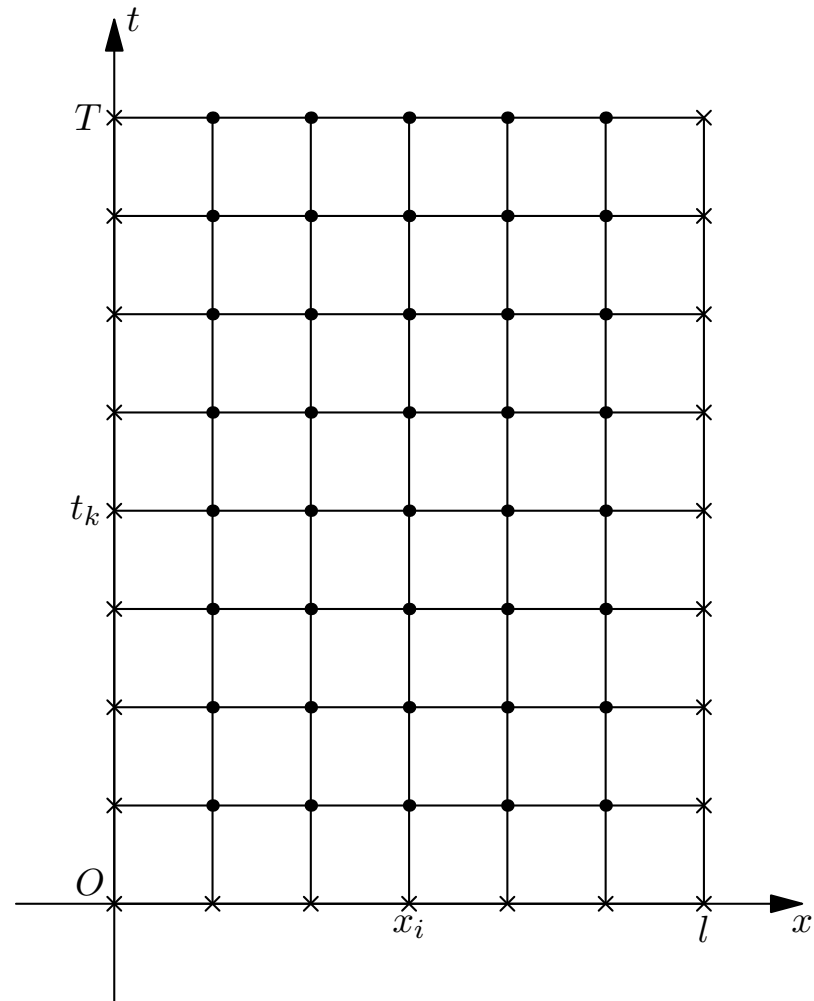


图 3.1 网格剖分

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_k) = \frac{1}{h^2} [u(\underline{x_{i+1}}, t_k) - 2u(\underline{x_i}, t_k) + u(\underline{x_{i-1}}, t_k)] - \frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\underline{\xi_i^k}, t_k), \quad \underline{\xi_i^k} \in (x_{i-1}, x_{i+1}),$$

上面 2 式代入 (3.1) 得

$$\begin{aligned} & \frac{1}{\tau^2} [u(x_i, t_{k+1}) - 2u(x_i, t_k) + u(x_i, t_{k-1}))] \\ & - \frac{a^2}{h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)] \\ & = f(x_i, t_k) + \underline{R_{ik}}, \quad 1 \leq i \leq M-1, \quad 1 \leq k \leq \underline{N-1}, \end{aligned}$$

≡ 局部误差

其中

$$R_{ik} = \frac{\tau^2}{12} \frac{\partial^4 u}{\partial t^4}(x_i, \eta_i^k) - \frac{a^2 h^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k).$$

1 ————— ?
0 —————

另外, 由初始条件, 有

$$\begin{aligned} u(x_i, 0) &= \varphi(x_i), \quad 1 \leq i \leq M-1, \\ u(x_i, t_1) &= u(x_i, 0) + \tau \frac{\partial u}{\partial t}(x_i, 0) + \frac{\tau^2}{2} \frac{\partial^2 u}{\partial t^2}(x_i, 0) + \frac{\tau^3}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i) \end{aligned}$$

$$\begin{aligned}
&= \varphi(x_i) + \tau\psi(x_i) + \frac{\tau^2}{2}[a^2\varphi''(x_i) + f(x_i, 0)] \\
&\quad + \frac{\tau^3}{6}\frac{\partial^3 u}{\partial t^3}(x_i, \eta_i) \\
&\equiv \Psi(x_i) + \frac{\tau^3}{6}\frac{\partial^3 u}{\partial t^3}(x_i, \eta), \quad 1 \leq i \leq M-1.
\end{aligned}$$

由边界条件, 有

$$u(0, t_k) = \alpha(t_k), \quad u(x_M, t_k) = \beta(t_k), \quad 0 \leq k \leq N.$$

忽略 R_{ik} 得下面的差分格式

$$\begin{cases} \frac{1}{\tau^2}(u_i^{k+1} - 2u_i^k + u_i^{k-1}) - \frac{a^2}{h^2}(u_{i+1}^k - 2u_i^k + u_{i-1}^k) = f(x_i, t_k), \\ \quad 1 \leq i \leq M-1, \quad 1 \leq k \leq N-1, \end{cases} \quad (3.2)$$

$$u_i^0 = \varphi(x_i), \quad u_i^1 = \Psi(x_i), \quad 1 \leq i \leq M-1, \quad (3.3)$$

$$u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \quad (3.4)$$

记 $s = a\tau/h$, s 称为步长比, 则上面的差分格式可以写成

$$\begin{aligned}
u_i^{k+1} &= s^2(u_{i+1}^k + u_{i-1}^k) + 2(1 - s^2)u_i^k - u_i^{k-1} + \tau^2 f(x_i, t_k), \\
&\quad 1 \leq i \leq M-1, \quad 1 \leq k \leq N-1.
\end{aligned}$$

例 3.1 考虑双曲方程初边值问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0, & 0 < x < 1, 0 < t \leq 1, \\ u(x, 0) = \mathbf{e}^x, \quad u_t(x, 0) = \mathbf{e}^x, & 0 < x < 1, \\ u(0, t) = \mathbf{e}^t, \quad u(1, t) = \mathbf{e}^{1+t}, & 0 < t \leq 1, \end{cases}$$

该问题的精确解为 $u(x, t) = \mathbf{e}^{x+t}$. 应用差分格式(3.2)-(3.4) 进行计算.

表 3.1 给出了 $h = 1/100$, $\tau = 1/100$ 时部分数值结果,数值解较好逼近精确解;表 3.2 给出了 $h = 1/100$, $\tau = 1/80$ 时部分数值结果,随着层数道的增加,误差也增加.

表 3.1 双曲方程显格式 $h = 1/100, \tau = 1/100$

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.0)	1.648721	1.648721	0.000000
10	(0.5, 0.1)	1.8222116	1.822119	0.000003
20	(0.5, 0.2)	2.013747	2.013753	0.000006
30	(0.5, 0.3)	2.225953	2.225541	0.000412
40	(0.5, 0.4)	2.459592	2.459603	0.000011
50	(0.5, 0.5)	2.718268	2.718282	0.000014
60	(0.5, 0.6)	3.004154	3.004166	0.000012
70	(0.5, 0.7)	3.320108	3.320117	0.000009
80	(0.5, 0.8)	3.669291	3.669297	0.000006
90	(0.5, 0.9)	4.055199	4.055200	0.000001
100	(0.5, 1.0)	4.481688	4.481689	0.000001

表 3.2 双曲方程显格式 $h = 1/100, \tau = 1/80$

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.0)	1.648721	1.648721	0.000000
1	(0.5, 0.0125)	1.669459	1.699460	0.000001
2	(0.5, 0.0250)	1.690458	1.690459	0.000001
3	(0.5, 0.0375)	1.711720	1.711722	0.000002
4	(0.5, 0.050)	1.733254	1.733253	0.000001
5	(0.5, 0.0625)	1.755042	1.755055	0.000013
6	(0.5, 0.0750)	1.777168	1.777130	0.000038
7	(0.5, 0.0875)	1.799323	1.799484	0.000161
8	(0.5, 0.100)	1.822730	1.822119	0.000611
9	(0.5, 0.1125)	1.842627	1.845038	0.002411
10	(0.5, 0.1250)	1.877649	1.868246	0.009403
11	(0.5, 0.1375)	1.854973	1.891746	0.036773
12	(0.5, 0.1500)	2.059241	1.915541	0.143700
13	(0.5, 0.1625)	1.377665	1.939635	0.561970
14	(0.5, 0.1750)	4.163094	1.964033	2.199061
15	(0.5, 0.1875)	-6.623787	1.988737	8.612524
16	(0.5, 0.200)	35.774890	2.013753	33.761137

关于显式差分格式 (3.2)-(3.4), 有如下结论:

定理 3.1 (1) 当步长比 $s \leq 1$ 时, 显式差分格式 (3.2)-(3.4) 在 L_2 范数下是稳定的; 当步长比 $s > 1$ 时, 显式差分格式 (3.2)-(3.4) 在 L_2 范数下是不稳定的;

(2) 当步长比 $s \leq 1$ 时, 显式差分格式 (3.2)-(3.4) 在 L_2 范数下关于空间步长和时间步长都是二阶收敛的。

3.2 隐格式

还是考虑 (x_i, t_k) 点的方程

$$\frac{\partial^2 u}{\partial t^2}(x_i, t_k) - a^2 \frac{\partial^2 u}{\partial x^2}(x_i, t_k) = f(x_i, t_k).$$

利用平均公式得

$$\frac{\partial^2 u}{\partial t^2}(x_i, t_k) - \frac{a^2}{2} \left[\frac{\partial^2 u}{\partial x^2}(x_i, t_{k+1}) + \frac{\partial^2 u}{\partial x^2}(x_i, t_{k-1}) \right] \quad (3.5)$$

$$= f(x_i, t_k) - \frac{a^2 \tau^2}{2} \frac{\partial^4 u}{\partial x^2 \partial t^2}(x_i, \bar{\eta}_i^k), \quad (3.6)$$

将

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2}(x_i, t_k) = & \frac{1}{\tau^2} [u(x_i, t_{k+1}) - 2u(x_i, t_k) + u(x_i, t_{k-1})] \\ & - \frac{\tau^2}{12} \frac{\partial^4 u}{\partial t^4}(x_i, \eta_i^k), \quad \eta_i^k \in (t_{k-1}, t_{k+1}), \end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_k) = \frac{1}{h^2} [u(x_{i+1}, t_k) - 2u(x_i, t_k) + u(x_{i-1}, t_k)]$$

$$-\frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i^k, t_k), \quad \xi_i \in (x_{i-1}, x_{i+1}),$$

代入 (3.6) 得

$$\begin{aligned} & \frac{1}{\tau^2} [u(x_i, t_{k+1}) - 2u(x_i, t_k) + u(x_i, t_{k-1})] - \frac{a^2}{2h^2} [u(x_{i+1}, t_{k+1}) \\ & \quad - 2u(x_i, t_{k+1}) + u(x_{i-1}, t_{k+1}) + u(x_{i+1}, t_{k-1}) \\ & \quad - 2u(x_i, t_{k-1}) + u(x_{i-1}, t_{k-1})] \\ & = f(x_i, t_k) + R_{ik}, \quad 1 \leq i \leq M-1, \quad 1 \leq k \leq N-1, \end{aligned}$$

其中

$$\begin{aligned} R_{ik} = & -\frac{a^2 \tau^2}{2} \frac{\partial^4 u}{\partial x^2 \partial t^2}(x_i, \bar{\eta}_i^k) + \frac{\tau^2}{12} \frac{\partial^4 u}{\partial t^4}(x_i, \eta_i^k) \\ & - \frac{a^2 h^2}{24} \left[\frac{\partial^4 u}{\partial x^4}(\bar{\xi}_i^{k+1}, t_{k+1}) + \frac{\partial^4 u}{\partial x^4}(\bar{\xi}_i^{k-1}, t_{k-1}) \right] \end{aligned}$$

由初边值条件

$$\begin{aligned} u(x_i, t_0) &= \varphi(x_i), \quad u(x_i, t_1) = \Psi(x_i) + \frac{\tau^3}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i), \\ & 1 \leq i \leq M-1, \end{aligned}$$

$$u(x_0, t_k) = \alpha(t_k), \quad u(x_M, t_k) = \beta(t_k), \quad 0 \leq k \leq N.$$

忽略上面的 R_{ik} 和 $\frac{\tau^3}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \eta_i)$, 并用 u_i^k 代替 $u(x_i, t_k)$ 的下面的差分格式

$$\left\{ \begin{array}{l} \frac{1}{\tau^2}(u_i^{k+1} - 2u_i^k + u_i^{k-1}) - \frac{a^2}{2h^2}(u_{i+1}^{k+1} - 2u_i^{k+1} + u_{i-1}^{k+1} \\ \quad + u_{i+1}^{k-1} - 2u_i^{k-1} + u_{i-1}^{k-1}) = f(x_i, t_k), \\ \quad \quad \quad 1 \leq i \leq M-1, \quad 1 \leq k \leq N-1, \\ u_i^0 = \varphi(x_i), \quad u_i^1 = \Psi(x_i), \quad 1 \leq i \leq M-1, \\ u_0^k = \alpha(t_k), \quad u_M^k = \beta(t_k), \quad 0 \leq k \leq N. \end{array} \right. \quad (3.7)$$

$$(3.8)$$

$$(3.9)$$

将 (3.7)–(3.9) 写成向量和矩阵形式

$$\begin{bmatrix} 1+s^2 & -\frac{1}{2}s^2 & & & \\ -\frac{1}{2}s^2 & 1+s^2 & -\frac{1}{2}s^2 & & \\ & \ddots & \ddots & \ddots & \\ & & -\frac{1}{2}s^2 & 1+s^2 & -\frac{1}{2}s^2 \\ & & & -\frac{1}{2}s^2 & 1+s^2 \end{bmatrix} \begin{bmatrix} u_1^{k+1} \\ u_2^{k+1} \\ \vdots \\ u_{M-2}^{k+1} \\ u_{M-1}^{k+1} \end{bmatrix}$$

$$\begin{aligned}
&= 2 \begin{bmatrix} u_1^k \\ u_2^k \\ \vdots \\ u_{M-2}^k \\ u_{M-1}^k \end{bmatrix} + \begin{bmatrix} -(1+s^2) & \frac{1}{2}s^2 & & & \\ \frac{1}{2}s^2 & -(1+s^2) & \frac{1}{2}s^2 & & \\ & \ddots & \ddots & \ddots & \\ & & \frac{1}{2}s^2 & -(1+s^2) & \frac{1}{2}s^2 \\ & & & \frac{1}{2}s^2 & -(1+s^2) \end{bmatrix} \\
&\quad \begin{bmatrix} u_1^{k-1} \\ u_2^{k-1} \\ \vdots \\ u_{M-2}^{k-1} \\ u_{M-1}^{k-1} \end{bmatrix} + \begin{bmatrix} \tau^2 f(x_1, t_k) + \frac{1}{2}s^2(\alpha(t_{k-1}) + \alpha(t_{k+1})) \\ \tau^2 f(x_2, t_k) \\ \vdots \\ \tau^2 f(x_{M-2}, t_k) \\ \tau^2 f(x_{M-1}, t_k) + \frac{1}{2}s^2(\beta(t_{k-1}) + \beta(t_{k+1})) \end{bmatrix}, \\
&\quad 1 \leq k \leq N-1.
\end{aligned}$$

例 3.2 用隐格式(3.7)–(3.9)计算例3.1所给定解问题的近似解.

取 $h = 1/100$, $\tau = 1/80$, 所得部分结果列表 3.3, 数值结果很好逼近精确解.

表 3.3 双曲方程隐格式 $h = 1/100, \tau = 1/80$

k	(x, t)	数值解	精确解	误差
0	(0.5, 0.0)	1.648721	1.648721	0.000000
10	(0.5, 0.125)	1.868241	1.868246	0.000005
20	(0.5, 0.250)	2.116993	2.117000	0.000007
30	(0.5, 0.375)	2.398868	2.398875	0.000007
40	(0.5, 0.50)	2.718278	2.718282	0.000004
50	(0.5, 0.625)	3.080229	3.080217	0.000012
60	(0.5, 0.75)	3.490369	3.490343	0.000026
70	(0.5, 0.875)	3.955116	3.955077	0.000040
80	(0.5, 1.00)	4.481741	4.481689	0.000052

关于隐式差分格式 (3.7)-(3.9), 有如下结论:

定理 3.2 (1) 对任意步长比 s , 隐式差分格式 (3.7)-(3.9) 在 L_2 范数下是稳定的; (2) 对任意步长比 s , 隐式差分格式 (3.7)-(3.9) 在 L_2 范数下关于空间步长和时间步长都是二阶收敛的。

4 椭圆型方程的差分方法

考虑二维 Poisson 方程 Dirichlet 边值问题

$$\begin{cases} -\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = f(x, y), & (x, y) \in \Omega, \\ u = \varphi(x, y), & (x, y) \in \Gamma. \end{cases} \quad (4.1)$$

$$(4.2)$$

其中 Ω 为矩形区域

$$\Omega = \{(x, y) \mid a < x < b, c < y < d\},$$

Γ 是 Ω 的边界.

4.1 差分格式的建立

将区间 $[a, b]$ 作 m 等分, 将 $[c, d]$ 作 n 等分, 记 $h_1 = (b - a)/m$, $h_2 = (d - c)/n$, $x_i = a + ih_1$, $0 \leq i \leq m$, $y_j = c + jh_2$, $0 \leq j \leq n$. h_1, h_2 称为 x 方向和 y 方向的步长. 用平行线

$$\begin{aligned} x &= x_i, & 0 \leq i \leq m, \\ y &= y_j, & 0 \leq j \leq n \end{aligned}$$

将 Ω 剖分为 mn 个小矩形, 交点 (x_i, y_j) 称为网格节点. 记

$$\Omega_h = \left\{ (x_i, y_j) \mid 0 \leq i \leq m, 0 \leq j \leq n \right\},$$

$$\overset{o}{\Omega}_h = \left\{ (x_i, y_j) \mid 1 \leq i \leq m-1, 1 \leq j \leq n-1 \right\},$$

$$\Gamma_h = \Omega_h \setminus \overset{o}{\Omega}_h.$$

称 $\overset{o}{\Omega}_h$ 为内结点, 称 Γ_h 为边界结点.

为方便起见, 记

$$\omega \equiv \left\{ (i, j) \mid (x_i, y_j) \in \overset{o}{\Omega}_h \right\}, \quad \gamma \equiv \left\{ (i, j) \mid (x_i, y_j) \in \Gamma_h \right\}.$$

记

$S_h = \{v \mid v = \{v_{ij} \mid 0 \leq i, j \leq m\} \text{ 为 } \Omega_h \text{ 上的网格函数} \}.$

设 $v = \{v_{ij} \mid 0 \leq i \leq m, 0 \leq j \leq n\} \in S_h$, 引进如下记号:

$$D_x v_{ij} = \frac{1}{h_1} (v_{i+1,j} - v_{ij}),$$

$$D_{\bar{x}} v_{ij} = \frac{1}{h_1} (v_{i,j} - v_{i-1,j}),$$

$$D_y v_{ij} = \frac{1}{h_2} (v_{i,j+1} - v_{ij}),$$

$$D_{\bar{y}} v_{ij} = \frac{1}{h_2} (v_{i,j} - v_{i,j-1}),$$

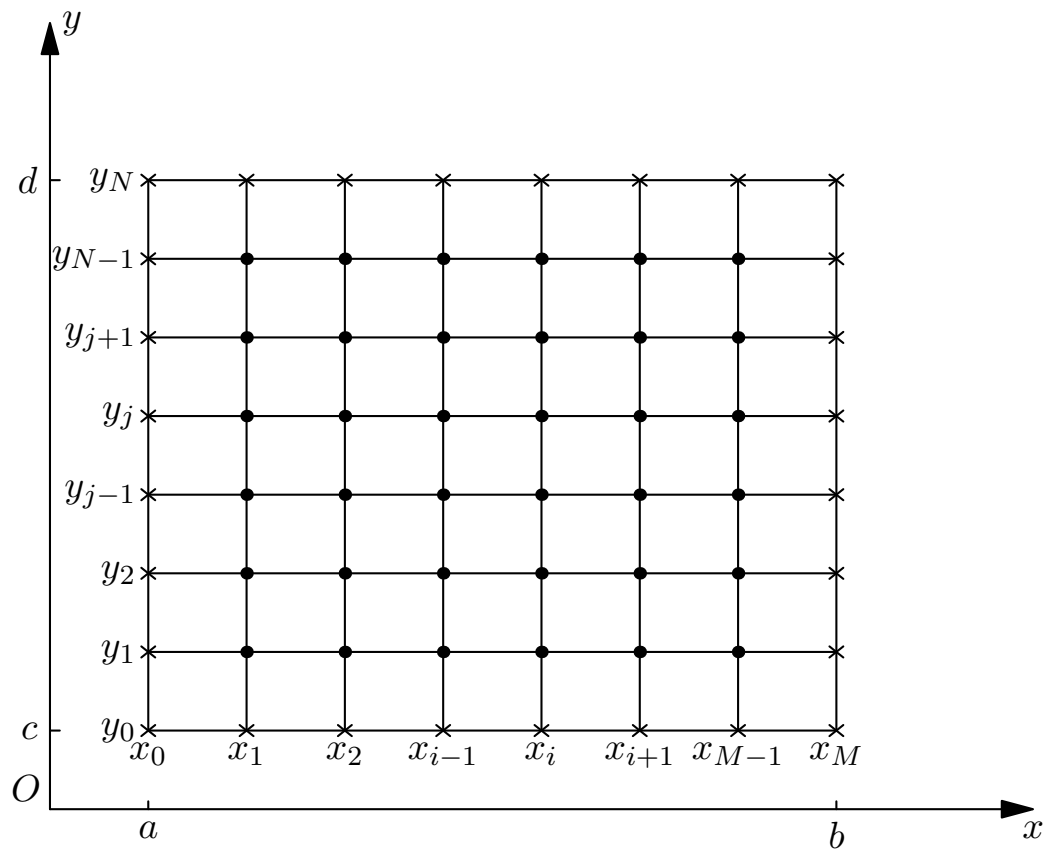


图 4.1 网格剖分

$$\delta_x^2 v_{ij} = \frac{1}{h_1}(D_x v_{ij} - D_{\bar{x}} v_{ij}), \quad \delta_y^2 v_{ij} = \frac{1}{h_2}(D_y v_{ij} - D_{\bar{y}} v_{ij}),$$

$$\|v\|_\infty = \max_{\substack{0 \leq i \leq m \\ 0 \leq j \leq n}} |v_{ij}|.$$

称 $\|v\|_\infty$ 为 v 的无穷范数.

在结点处考虑边值问题 (4.1)-(4.2), 有

$$-\left[\frac{\partial^2 u}{\partial x^2}(x_i, y_j) + \frac{\partial^2 u}{\partial y^2}(x_i, y_j)\right] = f(x_i, y_j), \quad (i, j) \in \omega, \quad (4.3)$$

$$u(x_i, y_j) = \varphi(x_i, y_j), \quad (i, j) \in \gamma. \quad (4.4)$$

定义 Ω_h 上的网格函数

$$U = \left\{ U_{ij} \mid 0 \leq i \leq m, 0 \leq j \leq n \right\},$$

其中

$$U_{ij} = u(x_i, y_j), \quad 0 \leq i \leq m, \quad 0 \leq j \leq n.$$

由公式 (0.4), 有

$$\frac{\partial^2 u}{\partial x^2}(x_i, y_j) = \frac{1}{h_1^2} \left[u(x_{i-1}, y_j) - 2u(x_i, y_j) + u(x_{i+1}, y_j) \right] - \frac{h_1^2}{12} \frac{\partial^4 u(\xi_{ij}, y_j)}{\partial x^4}$$

$$= \delta_x^2 U_{ij} - \frac{h_1^2}{12} \frac{\partial^4 u(\xi_{ij}, y_j)}{\partial x^4}, \quad x_{i-1} < \xi_{ij} < x_{i+1};$$

$$\frac{\partial^2 u}{\partial y^2}(x_i, y_j) = \frac{1}{h_2^2} \left[u(x_i, y_{j-1}) - 2u(x_i, y_j) + u(x_i, y_{j+1}) \right] - \frac{h_2^2}{12} \frac{\partial^4 u(x_i, \eta_{ij})}{\partial y^4}$$

$$= \delta_y^2 U_{ij} - \frac{h_2^2}{12} \frac{\partial^4 u(x_i, \eta_{ij})}{\partial y^4}, \quad y_{j-1} < \eta_{ij} < y_{j+1}.$$

将以上两式代入 (4.3), 并注意到边界条件 (4.4), 可得

$$-(\delta_x^2 U_{ij} + \delta_y^2 U_{ij}) = f(x_i, y_j) - \frac{h_1^2}{12} \frac{\partial^4 u(\xi_{ij}, y_j)}{\partial x^4} - \frac{h_2^2}{12} \frac{\partial^4 u(x_i, \eta_{ij})}{\partial y^4},$$

$$(i, j) \in \omega, \tag{4.5}$$

$$U_{ij} = \varphi(x_i, y_j), \quad (i, j) \in \gamma. \tag{4.6}$$

在上式中略去小量项

$$R_{ij} = -\frac{h_1^2}{12} \frac{\partial^4 u(\xi_{ij}, y_j)}{\partial x^4} - \frac{h_2^2}{12} \frac{\partial^4 u(x_i, \eta_{ij})}{\partial y^4}, \tag{4.7}$$

并用 u_{ij} 代替 U_{ij} , 得到如下差分格式(五点差分格式)

$$\begin{cases} -(\delta_x^2 u_{ij} + \delta_y^2 u_{ij}) = f(x_i, y_j), & (i, j) \in \omega, \\ u_{ij} = \varphi(x_i, y_j), & (i, j) \in \gamma. \end{cases} \quad (4.8)$$

称 R_{ij} 为差分格式 (4.8) 的局部截断误差, 它反映了差分格式 (4.8) 对精确解的满足程度, 即 R_{ij} 为差分格式 (4.8) 用精确解代替近似解后等式两边之差

$$R_{ij} = -\frac{1}{h_1^2} \left[u(x_{i-1}, y_j) - 2u(x_i, y_j) + u(x_{i+1}, y_j) \right] \\ - \frac{1}{h_2^2} \left[u(x_i, y_{j-1}) - 2u(x_i, y_j) + u(x_i, y_{j+1}) \right] - f(x_i, y_j).$$

记

$$M_4 = \max \left\{ \max_{(x,y) \in \bar{\Omega}} \left| \frac{\partial^4 u(x,y)}{\partial x^4} \right|, \max_{(x,y) \in \bar{\Omega}} \left| \frac{\partial^4 u(x,y)}{\partial y^4} \right| \right\}, \quad (4.10)$$

则有

$$|R_{ij}| \leq \frac{M_4}{12}(h_1^2 + h_2^2), \quad 1 \leq i \leq m-1, \quad 1 \leq j \leq n-1. \quad (4.11)$$

4.2 差分格式解的存在性

定理 4.1 差分格式 (4.8)-(4.9) 存在唯一解.

证明 差分格式 (4.8)-(4.9) 是线性的. 考虑其齐次方程组

$$\begin{cases} -(\delta_x^2 u_{ij} + \delta_y^2 u_{ij}) = 0, & (i, j) \in \omega, \end{cases} \quad (4.12)$$

$$\begin{cases} u_{ij} = 0, & (i, j) \in \gamma. \end{cases} \quad (4.13)$$

设 $\|u\|_\infty = M > 0$. 则由 (4.13) 知, 存在 $(i_0, j_0) \in \omega$ 使得 $|u_{i_0, j_0}| = M$, 且 $|u_{i_0-1, j_0}|, |u_{i_0+1, j_0}|, |u_{i_0, j_0-1}|, |u_{i_0, j_0+1}|$ 中至少有一个小于 M . 考虑 (4.12) 中 $(i, j) = (i_0, j_0)$ 的等式, 有

$$\left(\frac{2}{h_1^2} + \frac{2}{h_2^2}\right)u_{i_0, j_0} = \frac{1}{h_1^2}(u_{i_0-1, j_0} + u_{i_0+1, j_0}) + \frac{1}{h_2^2}(u_{i_0, j_0-1} + u_{i_0, j_0+1}).$$

将上式两边取绝对值, 可得

$$\begin{aligned}
 & \left(\frac{2}{h_1^2} + \frac{2}{h_2^2}\right)M \\
 & \leq \frac{1}{h_1^2}(|u_{i_0-1,j_0}| + |u_{i_0+1,j_0}|) + \frac{1}{h_2^2}(|u_{i_0,j_0-1}| + |u_{i_0,j_0+1}|) \\
 & < \left(\frac{2}{h_1^2} + \frac{2}{h_2^2}\right)M.
 \end{aligned}$$

与假设 $M > 0$ 矛盾. 故 $M = 0$. 因而差分格式 (4.8)-(4.9) 是唯一可解的. 定理证毕.

4.3 差分格式的求解

差分格式 (4.8)-(4.9) 是以 $\{u_{ij} \mid 1 \leq i \leq m-1, 1 \leq j \leq n-1\}$ 为未知量的线性方程组. (4.8) 可改写为

$$\begin{aligned}
 & -\frac{1}{h_2^2}u_{i,j-1} - \frac{1}{h_1^2}u_{i-1,j} + 2\left(\frac{1}{h_1^2} + \frac{1}{h_2^2}\right)u_{ij} - \frac{1}{h_1^2}u_{i+1,j} - \frac{1}{h_2^2}u_{i,j+1} \\
 & = f(x_i, y_j), \quad 1 \leq i \leq m-1, \quad 1 \leq j \leq n-1.
 \end{aligned} \tag{4.14}$$

记

$$\mathbf{u}_j = \begin{bmatrix} u_{1j} \\ u_{2j} \\ \vdots \\ u_{m-1,j} \end{bmatrix}, \quad 0 \leq j \leq n.$$

利用 (4.9) 可将 (4.14) 写为

$$\mathbf{D}\mathbf{u}_{j-1} + \mathbf{C}\mathbf{u}_j + \mathbf{D}\mathbf{u}_{j+1} = \mathbf{f}_j, \quad 1 \leq j \leq n-1, \quad (4.15)$$

其中

$$\mathbf{C} = \begin{bmatrix} 2\left(\frac{1}{h_1^2} + \frac{1}{h_2^2}\right) & -\frac{1}{h_1^2} & & & \\ -\frac{1}{h_1^2} & 2\left(\frac{1}{h_1^2} + \frac{1}{h_2^2}\right) & -\frac{1}{h_1^2} & & \\ & \ddots & \ddots & \ddots & \\ & & -\frac{1}{h_1^2} & 2\left(\frac{1}{h_1^2} + \frac{1}{h_2^2}\right) & -\frac{1}{h_1^2} \\ & & & -\frac{1}{h_1^2} & 2\left(\frac{1}{h_1^2} + \frac{1}{h_2^2}\right) \end{bmatrix},$$

$$D = \begin{bmatrix} -\frac{1}{h_2^2} & & & & \\ & -\frac{1}{h_2^2} & & & \\ & & \ddots & & \\ & & & -\frac{1}{h_2^2} & \\ & & & & -\frac{1}{h_2^2} \end{bmatrix}, \quad \mathbf{f}_j = \begin{bmatrix} f(x_1, y_j) + \frac{1}{h_1^2} \phi(x_0, y_j) \\ f(x_2, y_j) \\ \vdots \\ f(x_{m-2}, y_j) \\ f(x_{m-1}, y_j) + \frac{1}{h_1^2} \phi(x_m, y_j) \end{bmatrix}.$$

(4.15) 可进一步写为

$$\begin{bmatrix} C & D & & & \\ D & C & D & & \\ & \ddots & \ddots & \ddots & \\ & & D & C & D \\ & & & D & C \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \vdots \\ \mathbf{u}_{n-2} \\ \mathbf{u}_{n-1} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_1 - D\mathbf{u}_0 \\ \mathbf{f}_2 \\ \vdots \\ \mathbf{f}_{n-2} \\ \mathbf{f}_{n-1} - D\mathbf{u}_n \end{bmatrix}. \quad (4.16)$$

上述线性方程组的系数矩阵是一个三对角块矩阵, 每一行至多有 5 个非零元素, 通常称这种绝大多数元素为零的矩阵为**稀疏矩阵**. 常用迭代法求解以大型稀疏矩阵为系数矩阵的线性方程组. 可以证明 (4.16) 的系数矩阵是对称正定的.

Jacobi 迭代方法 对 $k = 0, 1, 2, \dots$, 计算

$$u_{ij}^{(k+1)} = \left[f(x_i, y_j) + \frac{1}{h_2^2} u_{i,j-1}^{(k)} + \frac{1}{h_1^2} u_{i-1,j}^{(k)} + \frac{1}{h_1^2} u_{i+1,j}^{(k)} + \frac{1}{h_2^2} u_{i,j+1}^{(k)} \right] / \left[2 \left(\frac{1}{h_1^2} + \frac{1}{h_2^2} \right) \right],$$
$$i = 1, 2, \dots, m-1; \quad j = 1, 2, \dots, n-1.$$

Gauss-Seidel 迭代方法 对 $k = 0, 1, 2, \dots$, 计算

$$u_{ij}^{(k+1)} = \left[f(x_i, y_j) + \frac{1}{h_2^2} u_{i,j-1}^{(k+1)} + \frac{1}{h_1^2} u_{i-1,j}^{(k+1)} + \frac{1}{h_1^2} u_{i+1,j}^{(k)} + \frac{1}{h_2^2} u_{i,j+1}^{(k)} \right] / \left[2 \left(\frac{1}{h_1^2} + \frac{1}{h_2^2} \right) \right],$$
$$i = 1, 2, \dots, m-1; \quad j = 1, 2, \dots, n-1.$$

例 4.1 应用差分格式 (4.8)-(4.9) 计算如下问题

$$\begin{cases} -\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = (\pi^2 - 1)e^x \sin(\pi y), & 0 < x < 2, 0 < y < 1, \\ u(0, y) = \sin(\pi y), \quad u(2, y) = e^2 \sin(\pi y), & 0 \leq y \leq 1, \\ u(x, 0) = 0, \quad u(x, 1) = 0, & 0 < x < 2. \end{cases}$$

该问题的精确解为 $u(x, y) = e^x \sin(\pi y)$.

将 $[0, 2]$ 作 m 等分, 将 $[0, 1]$ 作 n 等分, 用 Gauss-Seidel 迭代方法求解差分方程组 (4.8)-(4.9), 精确至 $\|u^{(l+1)} - u^{(l)}\|_\infty \leq \frac{1}{2} \times 10^{-10}$.

表 4.1 给出了 5 个结点处的精确解和取不同步长所得的数值解. 表 4.2 给出了这些结点处取不同步长时所得数值解和精确解差的绝对值 $|u(x_i, y_j) - u_{ij}|$. 表 4.3 给出了取不同步长时所得数值解的最大误差

$$E(h_1, h_2) = \max_{\substack{1 \leq i \leq m-1 \\ 1 \leq j \leq n-1}} |u(x_i, y_j) - u_{ij}|.$$

由表 4.3 可以看出, 当步长 h_1, h_2 同时缩小到原来的 $1/2$ 时, 最大误差约缩小到原来的 $1/4$.

表 4.3 取不同步长时数值解的最大误差

(h_1, h_2)	$E_\infty(h_1, h_2)$	$E_\infty(2h_1, 2h_2)/E_\infty(h_1, h_2)$
$(1/8, 1/8)$	$4.238e - 2$	*
$(1/16, 1/16)$	$1.061e - 2$	3.994
$(1/32, 1/32)$	$2.656e - 3$	3.995
$(1/64, 1/64)$	$6.640e - 4$	4.000

4.4 差分格式解的先验估计式

本节应用极值原理来给出差分方程组

$$\begin{cases} -(\delta_x^2 v_{ij} + \delta_y^2 v_{ij}) = g_{ij}, & (i, j) \in \omega, \end{cases} \quad (4.17)$$

$$\begin{cases} v_{ij} = \varphi_{ij}, & (i, j) \in \gamma \end{cases} \quad (4.18)$$

解的先验估计式.

设 $v = \{v_{ij} \mid 0 \leq i \leq m, 0 \leq j \leq n\}$ 为 Ω_h 上的网格函数, 记

$$(L_h v)_{ij} = -(\delta_x^2 v_{ij} + \delta_y^2 v_{ij}), \quad (i, j) \in \omega.$$

引理 4.1 (极值原理) 设 $v = \{v_{ij} \mid 0 \leq i \leq m, 0 \leq j \leq n\}$ 为 Ω_h 上的网格函数. 如果

$$(L_h v)_{ij} \leq 0, \quad (i, j) \in \omega,$$

则有

$$\max_{(i,j) \in \omega} v_{ij} \leq \max_{(i,j) \in \gamma} v_{ij}.$$

证明 用反证法. 设

$$\max_{(i,j) \in \omega} v_{ij} > \max_{(i,j) \in \gamma} v_{ij},$$

且 $\max_{(i,j) \in \omega} v_{ij} = M$, 则一定存在 $(i_0, j_0) \in \omega$ 使得 $v_{i_0, j_0} = M$, 且 v_{i_0-1, j_0} , v_{i_0+1, j_0} , v_{i_0, j_0-1} 和 v_{i_0, j_0+1} 中至少有一个的值小于 M . 因此

$$\begin{aligned} (L_h v)_{i_0, j_0} &= 2 \left(\frac{1}{h_1^2} + \frac{1}{h_2^2} \right) v_{i_0, j_0} - \frac{1}{h_1^2} (v_{i_0-1, j_0} + v_{i_0+1, j_0}) \\ &\quad - \frac{1}{h_2^2} (v_{i_0, j_0-1} + v_{i_0, j_0+1}) \end{aligned}$$

$$> 2 \left(\frac{1}{h_1^2} + \frac{1}{h_2^2} \right) M - \frac{1}{h_1^2} (M + M) - \frac{1}{h_2^2} (M + M) = 0.$$

这与条件矛盾. 定理证毕.

定理 4.2 设 $\{v_{ij}\}$ 为 (4.17)-(4.18) 的解, 则有

$$\max_{(i,j) \in \omega} |v_{ij}| \leq \max_{(i,j) \in \gamma} |\varphi_{ij}| + \frac{1}{4} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] \max_{(i,j) \in \omega} |g_{ij}|.$$

上式中 $\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2$ 为 Ω 的外接圆半径的平方.

证明 记

$$C = \max_{(i,j) \in \omega} |g_{ij}|,$$

$$P(x, y) = \left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 - \left(x - \frac{a+b}{2} \right)^2 - \left(y - \frac{c+d}{2} \right)^2,$$

并定义 Ω_h 上的网格函数

$$w_{ij} = \frac{1}{4} C P(x_i, y_j), \quad (i, j) \in \omega \cup \gamma,$$

则有

$$w_{ij} \geq 0, \quad (i, j) \in \omega \cup \gamma,$$

$$(L_h w)_{ij} = C, \quad (i, j) \in \omega.$$

因而

$$L_h(\pm v - w)_{ij} = \pm(L_h v)_{ij} - (L_h w)_{ij} = \pm g_{ij} - C \leq 0, \quad (i, j) \in \omega.$$

由极值原理, 知

$$\max_{(i,j) \in \omega} (\pm v - w)_{ij} \leq \max_{(i,j) \in \gamma} (\pm v - w)_{ij} \leq \max_{(i,j) \in \gamma} |\pm v_{ij}| + \max_{(i,j) \in \gamma} (-w_{ij}) \leq \max_{(i,j) \in \gamma} |v_{ij}|,$$

于是

$$\begin{aligned} \max_{(i,j) \in \omega} (\pm v)_{ij} &= \max_{(i,j) \in \omega} (\pm v - w + w)_{ij} \\ &\leq \max_{(i,j) \in \omega} (\pm v - w)_{ij} + \max_{(i,j) \in \omega} w_{ij} \\ &\leq \max_{(i,j) \in \gamma} |v_{ij}| + \max_{(i,j) \in \omega} w_{ij} \\ &\leq \max_{(i,j) \in \gamma} |\varphi_{ij}| + \frac{1}{4} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] \max_{(i,j) \in \omega} |g_{ij}|. \end{aligned}$$

易知

$$\max_{(i,j) \in \omega} |v_{ij}| \leq \max_{(i,j) \in \gamma} |\varphi_{ij}| + \frac{1}{4} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] \max_{(i,j) \in \omega} |g_{ij}|.$$

定理证毕.

4.5 差分格式解的收敛性和稳定性

收敛性

定理 4.3 设 $\{u(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$ 是定解问题 (4.1)-(4.2) 的解, $\{u_{ij} \mid 0 \leq i \leq m, 0 \leq j \leq n\}$ 为差分格式 (4.8)-(4.9) 的解, 则有

$$\max_{(i,j) \in \omega} |u(x_i, y_j) - u_{ij}| \leq \frac{M_4}{48} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] (h_1^2 + h_2^2),$$

其中 M_4 由 (4.10) 定义.

证明 记

$$e_{ij} = u(x_i, y_j) - u_{ij}, \quad (i, j) \in \omega \cup \gamma,$$

将 (4.5)-(4.6) 分别与 (4.8)-(4.9) 相减, 得误差方程

$$-\left(\delta_x^2 e_{ij} + \delta_y^2 e_{ij}\right) = R_{ij}, \quad (i, j) \in \omega, \quad (4.19)$$

$$e_{ij} = 0, \quad (i, j) \in \gamma, \quad (4.20)$$

其中 R_{ij} 由 (4.7) 定义. 由 (4.11), 有

$$\max_{(i,j) \in \omega} |R_{ij}| \leq \frac{M_4}{12} (h_1^2 + h_2^2).$$

应用定理 2.2.3, 有

$$\begin{aligned} \max_{(i,j) \in \omega} |e_{ij}| &\leq \frac{1}{4} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] \max_{(i,j) \in \omega} |R_{ij}| \\ &\leq \frac{M_4}{48} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] (h_1^2 + h_2^2). \end{aligned}$$

定理证毕.

稳定性

假设在应用差分格式 (4.8)-(4.9) 时, 计算右端函数 $f(x_i, y_j)$ 有误差 f_{ij} , 计算边界值有误差 φ_{ij} . 设 $\{v_{ij}\}$ 为差分格式

$$-\left(\delta_x^2 v_{ij} + \delta_y^2 v_{ij}\right) = f(x_i, y_j) + f_{ij}, \quad (i, j) \in \omega, \quad (4.21)$$

$$v_{ij} = \varphi(x_i, y_j) + \varphi_{ij}, \quad (i, j) \in \gamma \quad (4.22)$$

的解. 记

$$\varepsilon_{ij} = v_{ij} - u_{ij}, \quad (i, j) \in \omega \cup \gamma.$$

将 (4.21)-(4.22) 与 (4.8)-(4.9) 相减, 得

$$-\left(\delta_x^2 \varepsilon_{ij} + \delta_y^2 \varepsilon_{ij}\right) = f_{ij}, \quad (i, j) \in \omega, \quad (4.23)$$

$$\varepsilon_{ij} = \varphi_{ij}, \quad (i, j) \in \gamma. \quad (4.24)$$

应用定理 2.2.3, 得到

$$\max_{(i,j) \in \omega} |\varepsilon_{ij}| \leq \max_{(i,j) \in \gamma} |\varphi_{ij}| + \frac{1}{4} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] \max_{(i,j) \in \omega} |f_{ij}|.$$

由上式可知, 当 $\max_{(i,j) \in \gamma} |\varphi_{ij}|$ 和 $\max_{(i,j) \in \omega} |f_{ij}|$ 为小量时, $\max_{(i,j) \in \omega} |\varepsilon_{ij}|$ 也为小量. 我们称差分格式 (4.8) 和 (4.9) 关于边界值和右端函数是稳定的.

称 (4.23)-(4.24) 为摄动方程, 它的形式和差分格式 (4.8)-(4.9) 是一样的. 于是我们得到如下定理.

定理 4.4 差分格式 (4.8)-(4.9) 在下述意义下关于边界值和右端函数是稳定的: 设 $\{u_{ij}\}$ 为

$$\begin{aligned} -\left(\delta_x^2 u_{ij} + \delta_y^2 u_{ij}\right) &= f_{ij}, & (i, j) \in \omega, \\ u_{ij} &= \varphi_{ij}, & (i, j) \in \gamma \end{aligned}$$

的解, 则有

$$\max_{(i,j) \in \omega} |u_{ij}| \leq \max_{(i,j) \in \gamma} |\varphi_{ij}| + \frac{1}{4} \left[\left(\frac{b-a}{2} \right)^2 + \left(\frac{d-c}{2} \right)^2 \right] \max_{(i,j) \in \omega} |f_{ij}|.$$

$$a=1 \quad r = \frac{a\tau}{h^2}$$

$$b \frac{\partial u}{\partial x}$$

5 习题

p.345 ~ 346

$G-N$

1, 2, 3, 4, 7, 9, 10 (上机题)

12, 26 前支.

4, 5

1, 3 前支.